

Occupied Frames, Choi Covariances, and Effective Transfer Matrices for Gain–Loss Strings

Abid H. Bhuiyan
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This note gives a frame-level derivation of effective particle and hole transfer matrices for Choi states generated by ordered strings of number-conserving Gaussian operators, gains, and losses. The construction begins with the one-particle basis and the covariance-space convention for fermion modes, then rewrites fixed-charge Gaussian states as Slater occupied frames. Gain and loss operators act by adding or removing one direction from the occupied frame. In doubled Choi space the same operation propagates the occupied projector of the Choi state itself. Effective particle and hole transfer matrices are then graph coordinates of the final occupied or empty Choi subspace, with exact occupation caps appearing as zeros and their regulated complements as reciprocal poles.

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I. ONE-PARTICLE BASIS AND COVARIANCE-SPACE MODES

Let $\mathcal{H} \simeq \mathbb{C}^N$ be the one-particle Hilbert space with fixed orthonormal basis

$$\{e_i\}_{i=1}^N, \quad e_i^\dagger e_j = \delta_{ij}. \quad (1)$$

The fermion operators $\hat{c}_i, \hat{c}_i^\dagger$ obey

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij}, \quad \{\hat{c}_i, \hat{c}_j\} = 0, \quad \{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} = 0. \quad (2)$$

Throughout this note a one-particle column vector $x \in \mathcal{H}$ labels the covariance-space creation and annihilation modes

$$\hat{a}^\dagger(x) = \sum_i x_i^* \hat{c}_i^\dagger, \quad \hat{a}(x) = \sum_i x_i \hat{c}_i. \quad (3)$$

The complex conjugation in $\hat{a}^\dagger(x)$ is part of the convention. It gives

$$\{\hat{a}(\ell), \hat{a}^\dagger(g)\} = g^\dagger \ell. \quad (4)$$

Thus the same column x labels the one-particle state $\hat{a}^\dagger(x)|\text{vac}\rangle$ and its rank-one one-body density matrix $xx^\dagger$.

For a nonzero vector x , define the orthogonal projector

$$P[x] = x(x^\dagger x)^{-1}x^\dagger, \quad Q[x] = \mathbb{K} - P[x]. \quad (5)$$

For a full-rank frame $X = (x_1, \dots, x_r)$, define

$$P[X] = X(X^\dagger X)^{-1}X^\dagger, \quad Q[X] = \mathbb{K} - P[X]. \quad (6)$$

This formula is independent of the chosen basis of the subspace $\text{Ran } X$: if M is invertible, then $P[XM] = P[X]$.

II. FIXED-CHARGE GAUSSIAN STATES AS OCCUPIED FRAMES

Definition 1 (Occupied frame). *A charge- q Slater state is represented by an $N \times q$ frame $U = (u_1, \dots, u_q)$ through*

$$|U\rangle = \hat{a}^\dagger(u_1) \cdots \hat{a}^\dagger(u_q)|\text{vac}\rangle. \quad (7)$$

The occupied one-particle subspace is $\text{Ran } U$, and its projector is

$$C = P[U] = U(U^\dagger U)^{-1}U^\dagger. \quad (8)$$

If $U \mapsto UM$ with $M \in GL(q, \mathbb{C})$, the subspace and projector do not change, while

$$|UM\rangle = \det(M)^* |U\rangle. \quad (9)$$

The physical normalized Gaussian state is therefore the occupied subspace, whereas the frame stores a choice of basis and normalization.

The hole subspace and complex covariance are

$$\mathcal{H}_{\text{occ}} = \text{Ran } C, \quad \mathcal{H}_{\text{hole}} = \text{Ker } C = \text{Ran}(\mathbb{K} - C), \quad (10)$$

$$G = 2C - \mathbb{K}. \quad (11)$$

For a pure number-conserving Gaussian state,

$$C^\dagger = C, \quad C^2 = C, \quad G^\dagger = G, \quad G^2 = \mathbb{K}. \quad (12)$$

A. State-Covariance Identities

The fundamental Pauli constraints are

$$\hat{a}^\dagger(Cx)|U\rangle = 0, \quad \hat{a}((\mathbb{K} - C)x)|U\rangle = 0. \quad (13)$$

Creation in an already occupied direction vanishes; annihilation in an empty direction vanishes. In the local basis these become

$$\hat{c}_i^\dagger|G\rangle = -\sum_j G_{ij}\hat{c}_j^\dagger|G\rangle, \quad (14)$$

$$\hat{c}_i|G\rangle = \sum_j G_{ji}\hat{c}_j|G\rangle. \quad (15)$$

Equivalently,

$$(\mathbb{K} + G)\hat{c}^\dagger|G\rangle = 0, \quad (\mathbb{K} - G^T)\hat{c}|G\rangle = 0. \quad (16)$$

Thus the familiar covariance identities are not separate assumptions: they are the occupied-frame Pauli constraints in component form.

III. CHRONOLOGICAL FRAME PROPAGATION

Let $C = P[U]$ be the current occupied projector. The string is processed in chronological order: the rightmost operator in a product acts first.

A. Background State Updates

Gain. For a gain $\hat{a}^\dagger(g)$, only the empty component can be added:

$$h = (\mathbb{K} - C)g. \quad (17)$$

If $h = 0$, the state vanishes. Otherwise

$$C \mapsto C + P[h], \quad (18)$$

or, at the frame level, append h and re-orthonormalize if desired.

Loss. For a loss $\hat{a}(\ell)$, only the occupied component can be removed:

$$u = C\ell. \quad (19)$$

If $u = 0$, the state vanishes. Otherwise

$$C \mapsto C - P[u] = C - \frac{C\ell\ell^\dagger C}{\ell^\dagger C\ell}. \quad (20)$$

Geometrically, the new occupied subspace is

$$\text{Ran } C \cap \ell^\perp. \quad (21)$$

Even Gaussian. Let K be an invertible number-conserving Gaussian operator with one-particle occupied-frame map R :

$$K \hat{a}^\dagger(v) K^{-1} = \hat{a}^\dagger(Rv). \quad (22)$$

Then

$$U \mapsto RU, \quad C \mapsto P[RU]. \quad (23)$$

The associated annihilation-frame map is $R^{-\dagger}$.

B. Particle and Hole Tangents

A particle tangent over the background C is an extra vector $r \in \text{Ran}(\mathbb{K} - C)$, representing $\hat{a}^\dagger(r)|U\rangle$. After every update it is convenient to choose the representative orthogonal to the updated occupied subspace.

For an even K , with updated background C' ,

$$r \mapsto (\mathbb{K} - C')Rr. \quad (24)$$

For a gain, after adding $h = (\mathbb{K} - C)g$,

$$r \mapsto (\mathbb{K} - P[h])r. \quad (25)$$

For a loss, with $u = C\ell$,

$$r \mapsto r - u \frac{\ell^\dagger r}{\ell^\dagger u}. \quad (26)$$

Equation (26) is an oblique projection: it subtracts the removed occupied direction u so that the resulting vector is orthogonal to the loss wavefunction ℓ .

The hole algorithm is the mirror construction on the empty projector $E = \mathbb{K} - C$. A hole tangent z represents $\hat{a}(z)|U\rangle$ and is kept in the occupied complement of the empty frame. Under an even Gaussian the hole wavefunction transforms by $R^{-\dagger}$. A loss adds an empty cap, while a gain removes an empty direction obliquely. In words,

$$\text{particle: gains add caps, losses obliquely project,} \quad (27)$$

$$\text{hole: losses add caps, gains obliquely project.} \quad (28)$$

IV. CHOI OCCUPIED FRAMES

A. Doubled Space and Reference Frame

The Choi one-particle space is

$$\mathcal{H}_\Sigma = \mathcal{H}_L \oplus \mathcal{H}_R. \quad (29)$$

A doubled vector is written as

$$\begin{pmatrix} x_L \\ x_R \end{pmatrix}. \quad (30)$$

The number-conserving reference state is represented, up to an irrelevant normalization, by the occupied frame

$$F_{\text{REF}} = \begin{pmatrix} \mathbb{K} \\ \mathbb{K} \end{pmatrix}. \quad (31)$$

The normalized projector would be

$$C_{\text{REF}} = \frac{1}{2} \begin{pmatrix} \mathbb{K} & \mathbb{K} \\ \mathbb{K} & \mathbb{K} \end{pmatrix}, \quad (32)$$

but the unnormalized frame in Eq. (31) gives the same subspace.

For a right-leg string S_R , define

$$|\Sigma_S\rangle = S_R |\text{REF}\rangle. \quad (33)$$

If the final occupied frame is

$$F = \begin{pmatrix} A \\ B \end{pmatrix}, \quad (34)$$

then the Choi one-body projector and covariance are

$$C_\Sigma = F(F^\dagger F)^{-1} F^\dagger, \quad \Sigma = 2C_\Sigma - \mathbb{K}_{2N}. \quad (35)$$

B. Right-Leg Updates

A right-leg gain $\hat{a}_R^\dagger(g)$ is embedded as

$$\tilde{g} = \begin{pmatrix} 0 \\ g \end{pmatrix}. \quad (36)$$

It appends the empty component

$$h = (\mathbb{K} - C_\Sigma)\tilde{g} \quad (37)$$

provided $h \neq 0$.

A right-leg loss $\hat{a}_R(\ell)$ is embedded as

$$\tilde{\ell} = \begin{pmatrix} 0 \\ \ell \end{pmatrix}. \quad (38)$$

At the subspace level, the loss keeps

$$\text{Ran } F \mapsto \text{Ran } F \cap \tilde{\ell}^\perp. \quad (39)$$

Equivalently, if $m = \tilde{\ell}^\dagger F$, choose any matrix K whose columns span $\text{Ker } m$ and set $F \mapsto FK$. The choice-free projector update is

$$C_\Sigma \mapsto C_\Sigma - \frac{C_\Sigma \tilde{\ell} \tilde{\ell}^\dagger C_\Sigma}{\tilde{\ell}^\dagger C_\Sigma \tilde{\ell}}. \quad (40)$$

A right-leg even Gaussian K_R with occupied-frame map R acts by

$$F \mapsto \begin{pmatrix} \mathbb{K} & 0 \\ 0 & R \end{pmatrix} F. \quad (41)$$

V. TRANSFER MATRICES FROM CHOI FRAMES

A. Covariance Relation Pencils

Block the Choi covariance as

$$\Sigma = \begin{pmatrix} \Sigma_{LL} & \Sigma_{LR} \\ \Sigma_{RL} & \Sigma_{RR} \end{pmatrix}, \quad C_\Sigma = \begin{pmatrix} C_{LL} & C_{LR} \\ C_{RL} & C_{RR} \end{pmatrix}. \quad (42)$$

The doubled state identities give the particle and hole relation pencils

$$A_p \hat{c}_L^\dagger |\Sigma\rangle = -B_p \hat{c}_R^\dagger |\Sigma\rangle, \quad A_p = \mathbb{K} + \Sigma_{LL}, \quad B_p = \Sigma_{LR}, \quad (43)$$

$$A_h \hat{c}_L |\Sigma\rangle = B_h \hat{c}_R |\Sigma\rangle, \quad A_h = \mathbb{K} - \Sigma_{LL}^T, \quad B_h = \Sigma_{RL}^T. \quad (44)$$

When the relevant left block is invertible,

$$T_p = (\mathbb{K} + \Sigma_{LL})^{-1} \Sigma_{LR} = C_{LL}^{-1} C_{LR}, \quad (45)$$

$$T_h = (\mathbb{K} - \Sigma_{LL}^T)^{-1} \Sigma_{RL}^T = (\mathbb{K} - C_{LL})^{-T} C_{RL}^T. \quad (46)$$

If the inverse is singular, the pencil (A_p, B_p) or (A_h, B_h) remains the exact Choi-covariance data. A pseudoinverse is not a substitute for the missing source or cap sector.

B. Occupied Graph Coordinates and Particle Transfer

Let

$$F = \begin{pmatrix} A \\ B \end{pmatrix} \quad (47)$$

be a final occupied frame. The right-only occupied cap is

$$\mathcal{Y} = \{y \in \mathcal{H}_R : \begin{pmatrix} 0 \\ y \end{pmatrix} \in \text{Ran } F\}. \quad (48)$$

Algebraically, if Y is any frame for this subspace, then

$$\text{Ran } Y = B \text{ Ker } A. \quad (49)$$

When A has full row rank, choose any right inverse J satisfying

$$AJ = \mathbb{K}. \quad (50)$$

Then

$$FJ = \begin{pmatrix} \mathbb{K} \\ BJ \end{pmatrix}, \quad (51)$$

so $R_0 = BJ$ is a graph representative. It is not unique: changing J changes R_0 by a map into $\mathcal{Y}$. The covariance chart selects the canonical representative

$$R = Q[Y]R_0. \quad (52)$$

The particle transfer matrix in the Choi convention is

$$T_p = R^\dagger. \quad (53)$$

Equivalently, T_p is the unique representative of the particle relation with

$$T_p Y = 0. \quad (54)$$

The filled cap Y is not a choice; it is the right-only occupied part of the Choi subspace. Only the extension of the transfer matrix on that killed subspace is a coordinate choice.

C. Empty Graph Coordinates and Hole Transfer

Let $E = \begin{pmatrix} A_h \\ B_h \end{pmatrix}$ be a frame for the empty Choi subspace $\text{Ran}(\mathbb{K} - C_\Sigma)$. The right-only empty cap is

$$\mathcal{Z} = \{z \in \mathcal{H}_R : \begin{pmatrix} 0 \\ z \end{pmatrix} \in \text{Ran } E\}, \quad \text{Ran } Z = B_h \text{Ker } A_h. \quad (55)$$

If A_h has full row rank, choose J_h with $A_h J_h = \mathbb{K}$, set

$$H_0 = B_h J_h, \quad H = Q[Z]H_0. \quad (56)$$

For an empty graph $E \sim \begin{pmatrix} \mathbb{K} \\ H \end{pmatrix}$, Eq. (46) gives

$$T_h = -H^T. \quad (57)$$

The canonical hole representative kills the conjugated empty cap,

$$T_h \bar{Z} = 0. \quad (58)$$

The conjugation is a consequence of the hole convention in Eq. (44).

VI. REGULATED ENDCAPS

Suppose the finite particle chart has the form

$$T_p = N_p Q[Y], \quad (59)$$

where Y is the filled right-output cap. A simple endcap regulator is

$$Q[Y] \mapsto Q[Y] + \epsilon P[Y], \quad (60)$$

so

$$T_{p,\epsilon} = N_p (Q[Y] + \epsilon P[Y]). \quad (61)$$

The reciprocal hole representative is

$$T_{h,\epsilon} = T_{p,\epsilon}^{-T}. \quad (62)$$

Thus an exact zero in the particle chart appears as a reciprocal pole in the hole representative.

For a finite hole chart with empty cap Z ,

$$T_h = N_h Q[\bar{Z}], \quad (63)$$

the mirror regulator is

$$T_{h,\epsilon} = N_h (Q[\bar{Z}] + \epsilon P[\bar{Z}]), \quad T_{p,\epsilon} = T_{h,\epsilon}^{-T}. \quad (64)$$

The cap subspace is physical Choi-covariance data. The finite ϵ -extension is a regulated square-matrix representative.

VII. WORKED EXAMPLES

A. One Gain Between Evens: $K_{2,R} \hat{a}_R^\dagger(\chi) K_{1,R} | \text{REF} \rangle$

Let

$$K_i \hat{a}^\dagger(v) K_i^{-1} = \hat{a}^\dagger(R_i v), \quad T_i = R_i^\dagger. \quad (65)$$

Starting from F_{REF} , after $K_{1,R}$ one has

$$F_1 = \begin{pmatrix} \mathbb{K} \\ R_1 \end{pmatrix}. \quad (66)$$

The right gain appends $\begin{pmatrix} 0 \\ \chi \end{pmatrix}$, modulo the already occupied graph:

$$F_2 \sim \begin{pmatrix} \mathbb{K} & 0 \\ R_1 & \chi \end{pmatrix}. \quad (67)$$

After $K_{2,R}$,

$$F_f \sim \begin{pmatrix} \mathbb{K} & 0 \\ R_2 R_1 & R_2 \chi \end{pmatrix}. \quad (68)$$

Define

$$y = R_2 \chi = T_2^\dagger \chi, \quad Q_y = Q[y]. \quad (69)$$

The canonical occupied graph is

$$R_{\text{can}} = Q_y R_2 R_1. \quad (70)$$

Hence

$$T_p = R_{\text{can}}^\dagger = T_1 T_2 Q_y, \quad (71)$$

and the regulated particle and reciprocal hole representatives are

$$T_{p,\epsilon} = T_1 T_2 (Q_y + \epsilon P[y]), \quad T_{h,\epsilon} = T_{p,\epsilon}^{-T}. \quad (72)$$

There is a local Pauli projection at the gain insertion point, $Q[\chi]$, in the chronological particle tangent. It is redundant in the canonical final representative because

$$Q_y R_2 P[\chi] = 0. \quad (73)$$

B. Gain Followed by Loss: $K_{3,R} \hat{a}_R(\ell) K_{2,R} \hat{a}_R^\dagger(\chi) K_{1,R} | \text{REF} \rangle$

After K_1 , the gain, and K_2 , write

$$F_- = \begin{pmatrix} \mathbb{K} & 0 \\ A & y \end{pmatrix}, \quad A = R_2 R_1, \quad y = R_2 \chi. \quad (74)$$

A vector in $\text{Ran } F_-$ has the form

$$\begin{pmatrix} x \\ Ax + \alpha y \end{pmatrix}. \quad (75)$$

The right loss by ℓ imposes the condition

$$\begin{pmatrix} 0 \\ \ell \end{pmatrix}^\dagger \begin{pmatrix} x \\ Ax + \alpha y \end{pmatrix} = \ell^\dagger (Ax + \alpha y) = 0. \quad (76)$$

If

$$s = \ell^\dagger y \neq 0, \quad (77)$$

then

$$\alpha = -s^{-1} \ell^\dagger Ax. \quad (78)$$

Therefore

$$Ax + \alpha y = [\mathbb{K} - y(\ell^\dagger y)^{-1} \ell^\dagger] Ax. \quad (79)$$

The bracket is the oblique projector

$$O_{y|\ell} = \mathbb{K} - y(\ell^\dagger y)^{-1} \ell^\dagger, \quad O_{y|\ell} y = 0, \quad \ell^\dagger O_{y|\ell} = 0. \quad (80)$$

After $K_{3,R}$,

$$R_f = R_3 O_{y|\ell} R_2 R_1. \quad (81)$$

Thus

$$T_p = T_1 T_2 [\mathbb{K} - \ell(y^\dagger \ell)^{-1} y^\dagger] T_3. \quad (82)$$

The regulated form is

$$T_{p,\epsilon} = T_1 T_2 [\mathbb{K} - (1 - \epsilon) \ell(y^\dagger \ell)^{-1} y^\dagger] T_3, \quad T_{h,\epsilon} = T_{p,\epsilon}^{-T}. \quad (83)$$

No canonical assumption was used to determine α : it is fixed by the loss intersection condition $\text{Ran } F_- \cap (0 \oplus \ell)^\perp$.

C. Two Consecutive Gains: $K_{2,R} \hat{a}_R^\dagger(\chi_1) \hat{a}_R^\dagger(\chi_2) K_{1,R} | \text{REF} \rangle$

The rightmost gain $\chi_2^\dagger$ acts first. After $K_{2,R}$, the occupied frame can be written

$$F_f \sim \begin{pmatrix} \mathbb{K} & 0 & 0 \\ R_2 R_1 & R_2 \chi_2 & R_2 \chi_1 \end{pmatrix}. \quad (84)$$

Let

$$Y = R_2(\chi_2, \chi_1) = T_2^\dagger(\chi_2, \chi_1). \quad (85)$$

Assuming the two gain directions remain independent after projection, the finite particle representative is

$$T_p = T_1 T_2 Q[Y], \quad (86)$$

and the regulated pair is

$$T_{p,\epsilon} = T_1 T_2 (Q[Y] + \epsilon P[Y]), \quad T_{h,\epsilon} = T_{p,\epsilon}^{-T}. \quad (87)$$

D. Two Consecutive Losses: $K_{2,R} \hat{a}_R(\chi_2) \hat{a}_R(\chi_1) K_{1,R} | \text{REF} \rangle$

For losses the empty frame is the natural object. After $K_{1,R}$, an empty frame orthogonal to $\begin{pmatrix} \mathbb{K} \\ R_1 \end{pmatrix}$ is

$$E_1 = \begin{pmatrix} \mathbb{K} \\ -R_1^{-\dagger} \end{pmatrix}. \quad (88)$$

The two losses add right-only empty directions. After $K_{2,R}$, empty wavefunctions are propagated by $R_2^{-\dagger}$, giving

$$E_f \sim \begin{pmatrix} \mathbb{K} & 0 & 0 \\ -R_2^{-\dagger} R_1^{-\dagger} & R_2^{-\dagger} \chi_1 & R_2^{-\dagger} \chi_2 \end{pmatrix}. \quad (89)$$

Define the empty cap frame

$$Z = R_2^{-\dagger}(\chi_1, \chi_2). \quad (90)$$

The finite hole representative is

$$T_h = T_1^{-T} T_2^{-T} Q[\bar{Z}], \quad (91)$$

with regulated pair

$$T_{h,\epsilon} = T_1^{-T} T_2^{-T} (Q[\bar{Z}] + \epsilon P[\bar{Z}]), \quad T_{p,\epsilon} = T_{h,\epsilon}^{-T}. \quad (92)$$

E. A Two-Mode Sanity Check

Set all evens to the identity, choose $N = 2$, let

$$\chi = e_1, \quad \ell = e_1 + \alpha e_2. \quad (93)$$

For the gain-loss word $\hat{a}(\ell)\hat{a}^\dagger(\chi)$, one has

$$y = e_1, \quad y^\dagger \ell = 1, \quad (94)$$

and

$$\ell(y^\dagger \ell)^{-1} y^\dagger = \begin{pmatrix} 1 & 0 \\ \alpha & 0 \end{pmatrix}. \quad (95)$$

Equation (83) gives

$$T_{p,\epsilon} = \mathbb{K} - (1 - \epsilon)\ell y^\dagger = \begin{pmatrix} \epsilon & 0 \\ -(1 - \epsilon)\alpha & 1 \end{pmatrix}. \quad (96)$$

The finite limit

$$T_p = \begin{pmatrix} 0 & 0 \\ -\alpha & 1 \end{pmatrix} \quad (97)$$

has a particle zero in the filled direction, while

$$T_{h,\epsilon} = T_{p,\epsilon}^{-T} \quad (98)$$

has the reciprocal pole.

VIII. SUMMARY

The occupied-frame construction separates three pieces of data that are often conflated in transfer-matrix notation. First, the Choi covariance is the primary state-level object: $C_\Sigma = P[F]$. Second, the finite particle or hole transfer matrix is a graph coordinate of the occupied or empty Choi subspace. Third, a non-neutral string creates an exact right-output cap: a filled cap Y for net gain or an empty cap Z for net loss. The covariance chart returns the canonical representative that kills this cap, while the regulated endcap $Q + \epsilon P$ supplies an invertible square representative whose opposite particle-hole sector has a reciprocal pole.